

Assignment #2 – Objective

Workflow in R (or Python)

- Load spatial data (sf, terra)
- Subset and clean attributes
- Join external attribute table to spatial data
- Compute summary statistics
- Export results for visualization or modeling

Assignment #2 – Background

- Land parcels are boundaries that have associated information such as property owner, land use, value, and location attributes
- Improvement value refers to the assessed monetary value attributed not to the land itself but to the physical improvements on the land (typically buildings and other structures). In the context of property assessment, this is often used as a proxy for the built environment's contribution to total property worth (e.g., houses, garages, renovations).

Follows

1. Data Acquisition
2. Data Preparation
3. Aggregation and Analysis
4. Integrate Census Data
5. Visualizations
6. Documentation and Reporting



Assignment #2 – Part 1

1. Data Acquisition

Parcel Data

- Visit the Texas Natural Resources Information System (TNRIS) portal <https://tnris.org/stratmap/land-parcels.html>
- Select parcel data for 5 neighboring counties of your choice
- Download the 2025 parcel datasets for each county and unzip in a folder (a local folder with raw parcel shapefiles files for the counties)

Assignment #2 – Part 2

2. Data Preparation

Using R (with sf, dplyr, and tidyr)

a. Read the parcel files, standardize and inspect

- Combine all counties into a single sf object
- If needed, ensure projections^(and fields) are consistent across counties
- Inspect STAT_LAND_ and IMP_VALUE

b. Filter for single-family houses

- Define single family residential by the parcel state code:
STAT_LAND_ == "A"

Assignment #2 – Part 3

3. Aggregation and Analysis

a. Aggregate Improvement Value

- For each county, calculate:
 - The total improvement value of single-family parcels
 - Parcel count

Assignment #2 – Part 4

4. Integrate Census Data

a. Acquire Census TIGER/Line Data

- Use the `tidycensus` packages to download:
- County boundaries
- Population estimates (e.g., ACS 5-year estimate for total population)

b. Merge with Parcel Aggregates

- Join population data by county GEOID or name

c. Compute Per Capita Metrics

- Average improvement value per capita
- Average population per single-family parcel

Assignment #2 – Part 5

5. Visualization

a. Basic thematic maps

- Use `tmap` (or `ggplot2`) to produce:
 - Basic choropleth map of total improvement value by county
 - Basic choropleth map of average improvement value per inhabitant

b. You may also consider including supporting plots, such as:

- Bar charts or scatterplots showing:
 - Population vs. improvement value
 - Parcel count vs. per capita value

c. Interpretation

- Comment on spatial patterns
(e.g., regional real estate markets, population density relationships)

Assignment #2 – Part 6

6. Documentation and Reporting

a. Reproducible Script or Notebook

- Clean, commented (as possible)
- Reproducible from data download to map output

b. Include

- Data sources and preprocessing steps
- Results and interpretations
- Limitations and potential extensions (e.g., overlay zoning, flood risk)

c. Visuals

- Please include / submit maps and charts with captions